HW 5

- **Mirror plane** (m)

  Transformation matrix for m including x, y axises

$$\begin{pmatrix} x \\ y \\ -z \end{pmatrix} = \Delta \begin{pmatrix} x \\ y \\ z \end{pmatrix} \qquad \Delta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Exercise I:

Please give the transformation matrix when m include x、z axis ?

When m include y、z axis ?

m include x, z axis: $\Delta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ m include y, z axis: $\Delta = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Exercise:

Please derive all the transformation matrixes for L^6.

$$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\theta = 60°$ $\theta = 120°$ $\theta = 180°$ $\theta = 240°$ $\theta = 300°$ $\theta = 360°$

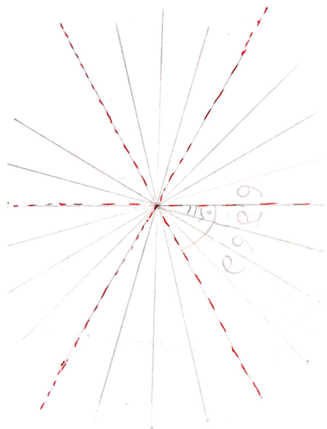
Exercise for Theorem 1:

$L^6 \times L^2_{\perp} \rightarrow ?$ **$L^6 6L^2$**

Exercise for converse theorem :

When L^2 intersects with L^2 with 15° , which symmetry axis can be generated?

Please give the symmetrical patterns to demonstrate the combination of symmetry elements above.

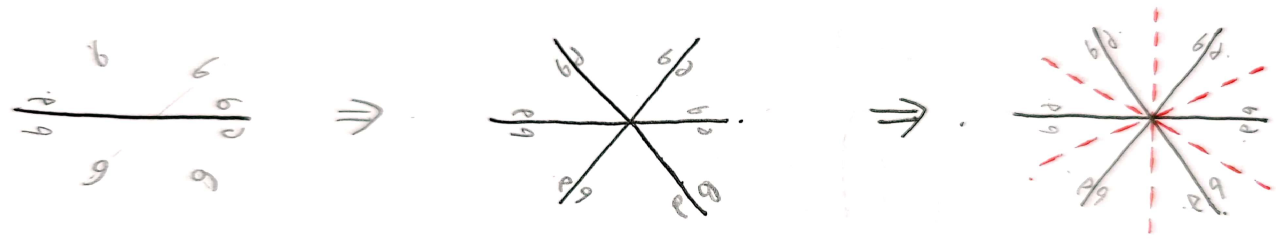


Exercise of theorem 3:

$$L^3 \times P_{//} \rightarrow ?$$

When two P intersect with 60° , which symmetry axis will be generated?

Please give the symmetrical patterns to demonstrate the combination of symmetry elements above.



Exercise: to build up a paper model of the above polyhedron.

Please post your photos here!

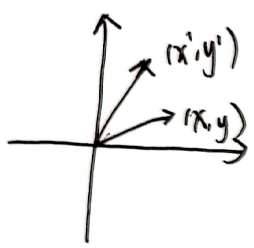
1. tetragonal tetrahedron

2. Zongzi

3. ditrigonal scalenohedron

Please prove transformation matrix of L^n

$$\begin{pmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



$x = r\cos\varphi$

$y = r\sin\varphi$

$x' = r\cos(\alpha+\varphi) = r\cos\alpha\cos\varphi - r\sin\alpha\sin\varphi = \cos\alpha\, x - \sin\alpha\, y$

$y' = r\sin(\alpha+\varphi) = r(\sin\alpha\cos\varphi + \sin\varphi\cos\alpha) = (\sin\alpha)x + (\cos\alpha)y$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

Please prove transformation matrix of L_i^n

$$\begin{pmatrix} -\cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & -\cos\alpha & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} \cos(-\alpha) & -\sin(-\alpha) & 0 \\ \sin(-\alpha) & \cos(-\alpha) & 0 \\ 0 & 0 & 1 \end{pmatrix}^T = \begin{pmatrix} -\cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & -\cos\alpha & 0 \\ 0 & 0 & -1 \end{pmatrix}$$